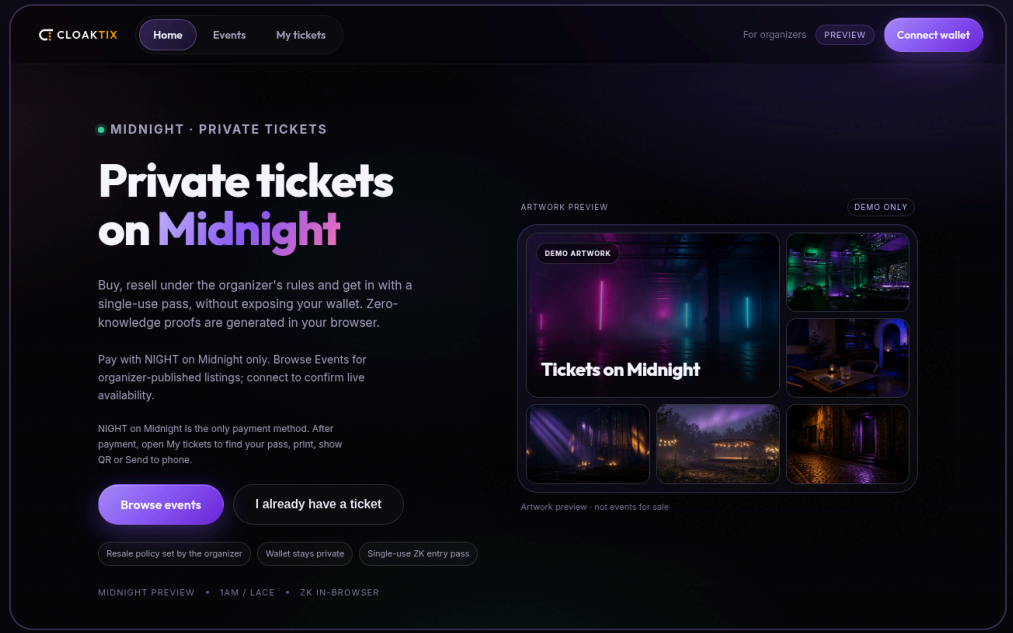
CLOAKTIX

Private, fair tickets on Midnight

Rules for resale. Privacy at the door.
Live on Midnight Preview, ledger 9.

Ledger 9 | NIGHT only | In-browser ZK proofs | 1AM wallet



Real capture of the current build

PROBLEM

Ticketing breaks trust twice.

Uncontrolled resale

Scalping puts fans and organizers at odds. Rules written in terms of service are not enough.

Enforce the resale policy on-chain.

Unnecessary identity exposure

Getting into an event should not mean handing over a wallet address or an identity document.

Prove eligibility in zero knowledge.

CloakTix combines **fair ticket rules** with **private proofs**, paid in NIGHT on Midnight.

PROGRESS DURING WAVE 3

Ledger 9 first, then what buyers asked for.

7 Nov

Ledger 9 migration

Compact 0.36, Midnight.js 5, ZKIR v3. Contract recompiled with new keys after the Preview reset.

9 Nov

Contract b92733 live

Deployed with 1AM. Schema 3, 3% fee, 13 circuits. Event 1 and a first ticket purchase on-chain.

9 Nov

In-tab ZKIR v3 prover

1AM proves v2 only, so the app proves its own circuits in a tab worker. 1AM still balances, pays DUST and submits.

9 Nov

Single-intent multi-buy

Fixed error 227 by merging every buyTicket into one intent. Accepted by 1AM live.

9 Nov

Cleaner buyer catalog

Key-registration entry hidden. Identical ticket types shown as one. Proof cache kept between visits.

9 Nov

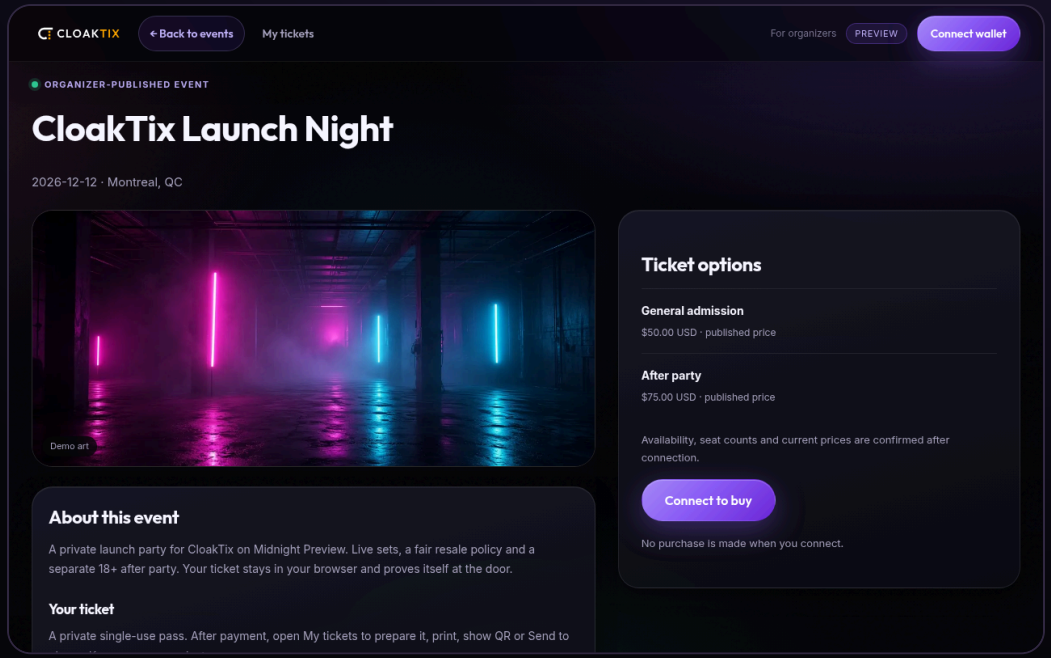
Age credential and labels

18+ types ask for a demo age credential first. Every ticket shows its type name.

DISCOVER WITHOUT A WALLET

Browse real listings before connecting.

- Organizer-published events: name, date, venue, description and cover, signed by the organizer.
- USD prices on every option. NIGHT is the only payment method.
- The key-registration entry (event 2) never shows to buyers.
- Two identical General admission types on event 1 show as one option.



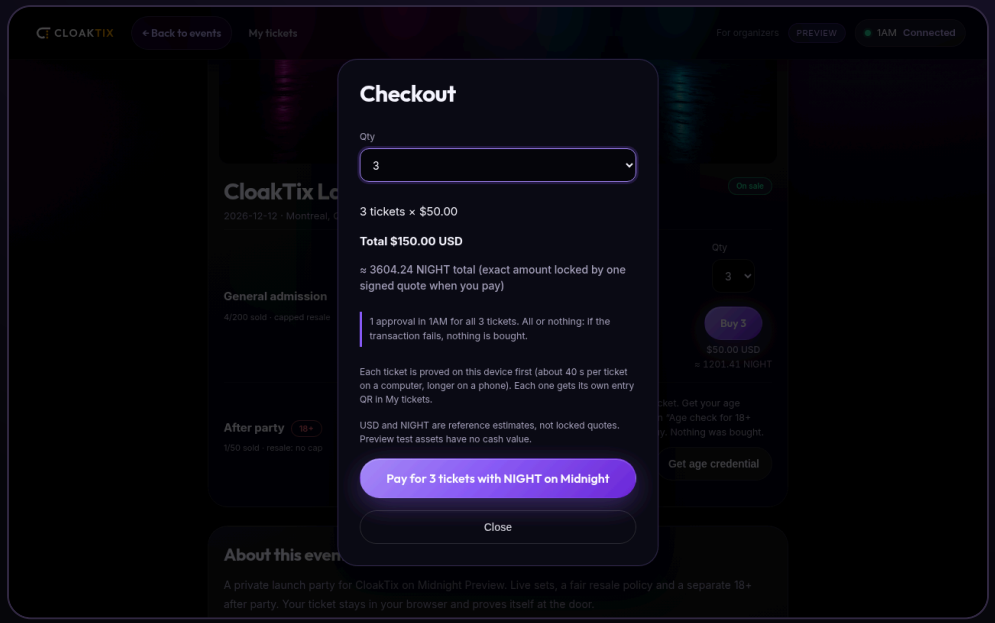
Real capture of the current build

Event text and cover on these frames are sample metadata in a local catalog.

MULTI-BUY, ONE APPROVAL

Up to 4 tickets in a single intent.

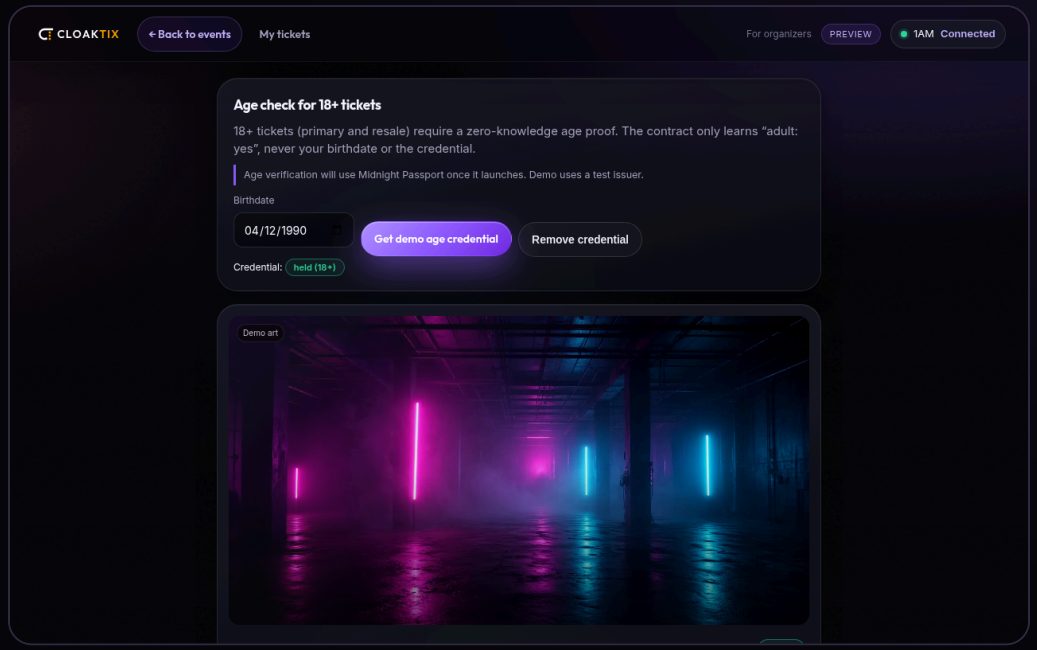
- Each ticket is proved on the device, then all buyTicket calls are merged into **one intent** before signing.
- 1AM sees one NIGHT input set and asks for **one approval**.
- All or nothing: if the transaction fails, nothing is bought.
- Fixes error 227 (two intents spending the same NIGHT coin). Confirmed live with 1AM on 9 Nov.



Real UI · live Preview contract state · read-only capture connector, nothing signed

18+ WITHOUT A BIRTHDATE

Prove "adult: yes". Nothing more.

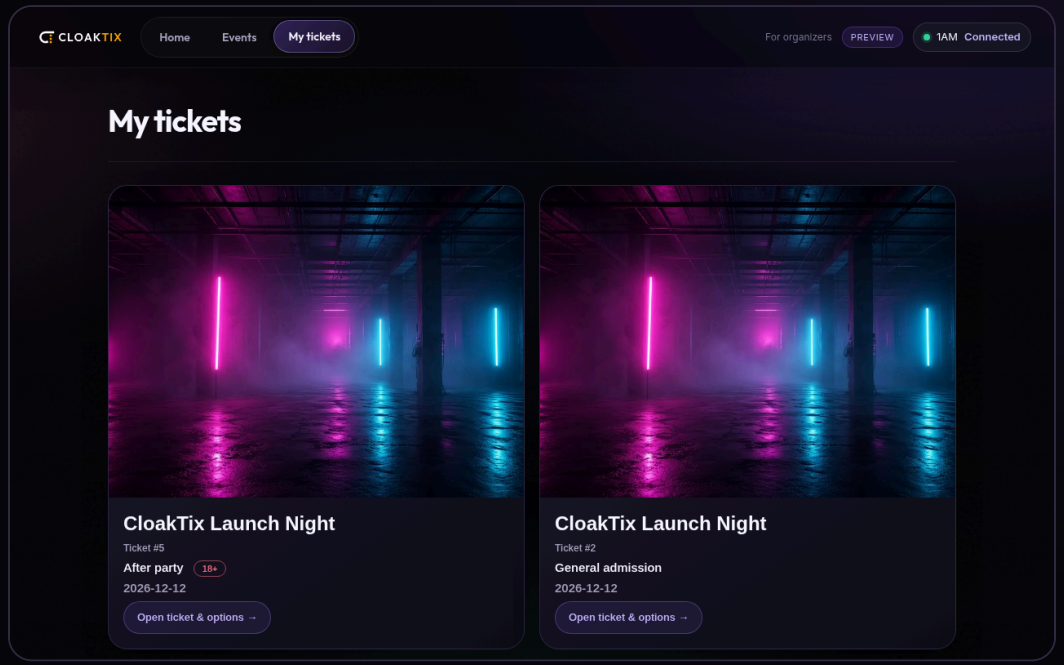


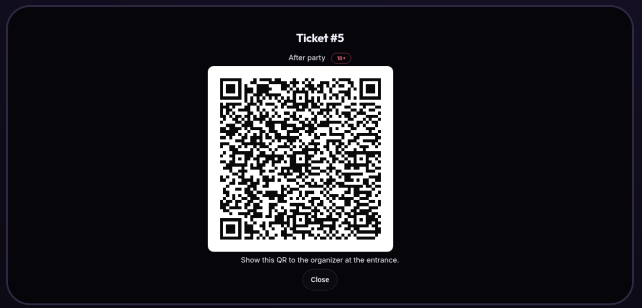
Real UI · live Preview contract state · read-only capture connector, nothing signed

- 18+ types show an age step first: **Get demo age credential**.
- buyTicketAdult checks the issuer signature in zero knowledge. The contract learns only "adult: yes".
- A missing or untrusted credential blocks Buy before any quote or proof.
- Demo "Passport stub" issuer today. A real issuer later is one adminSet, no new contract.

MY TICKETS

Every ticket shows its type.



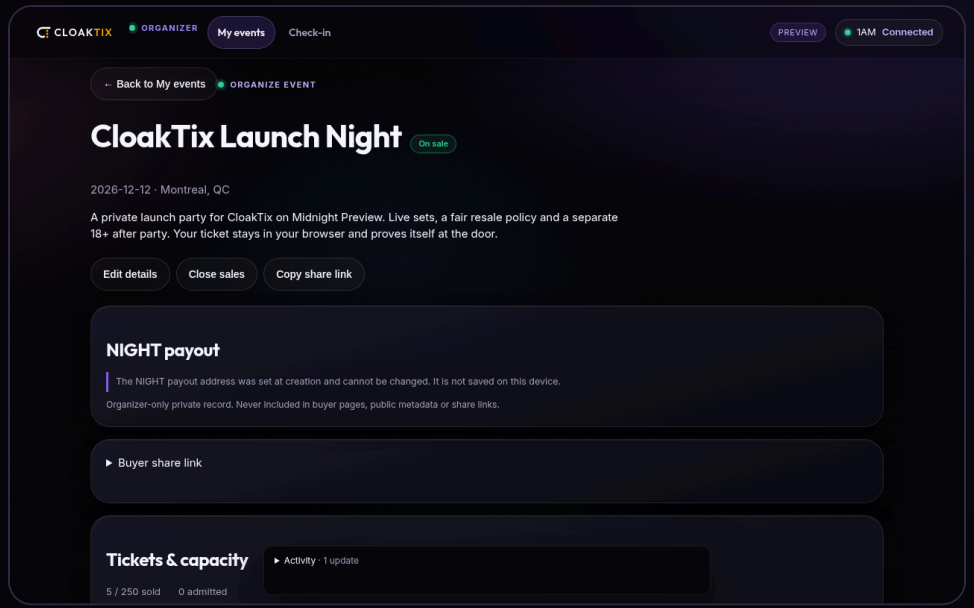


- Square-ish cards, two per row (one column on small phones).
- Ticket type name and 18+ badge on each card.
- Print, Show QR at the door, or Send to phone. Showing a ticket never performs a chain operation.

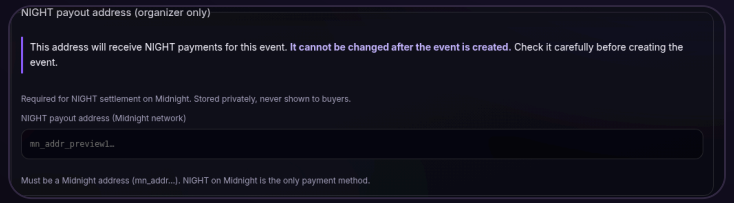
Sample data render · real components, sample ownership

ORGANIZER TOOLS

Run each event from its own page.



Sample data render · real components, sample ownership



Real capture of the current build

- Name, date, place, description and cover are required.
- The NIGHT payout address cannot be changed after creation, and the form says so.
- Sales, share link, ticket types and capacity. Duplicate type names are blocked.

RULES AND BUSINESS MODEL

Resale rules belong to the organizer.

Capped

Resale cannot exceed the organizer maximum.

No resale

Listing is refused. Entry still works.

Unlimited

No price cap. The fee still applies.

3%

On every sale and resale, set on-chain (feeBps 300).



Prices in USD, paid in NIGHT. No payout addresses shown.

ARCHITECTURE

Compact rules. In-tab proofs. 1AM.

Web app

React on a Cloudflare Worker

In-tab ZKIR v3 prover with a cached key store

Signed public catalog

→

1AM wallet

Balances the transaction

Pays DUST and submits

One approval per purchase

→

Compact contract

Schema 3, 13 circuits

Ticket policies, escrowed resale, entry

Age check via buyTicketAdult

→

Midnight Preview

Ledger 9

NIGHT payments

Built with OpenZeppelin Compact Contracts

One purchase: prove each ticket in the tab → merge into one intent → 1AM balances, pays DUST, submits → contract checks price, policy and age proof.

Contract `b92733389b1955c913dec004df708a12697df4a962ffab5dbde1e88f574ab04e`

STATUS AND LINKS

Working on Midnight Preview today.

Confirmed live with 1AM

Create event · ticket purchase · single-intent multi-buy · NIGHT checkout with USD prices · walletless catalog · Organize event · send to phone.

Code

github.com/peejay0786-star/cloaktix

Apache-2.0 · README explains how to run and test.

Honest limits

- Demo age issuer, not Midnight Passport yet.
- Resale and check-in not yet verified end to end on b92733.
- In-tab proving takes tens of seconds per ticket.
- NIGHT payments are unshielded (pseudonymous).
- Multi-door stays fail-closed. Prototype, not audited.